Natural Language Processing

(Spring 2026)

Assignment 1 Report

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Objective:

The main purpose of this Assignment 1 of the Natural Language Processing course is to understand and implement a complete pipeline of NLP, such as collecting text, cleaning it, extracting patterns using regex, tokenization, stopwords, stemming or lemmatization, POS, NER, building N-gram language models, unigram/bigram with and without smoothing and then finding perplexity. Implementing BPE for Urdu text and then building embedding models.

The assignment is divided into 4 equal parts with each has tasks in it.

PART 1

This part focuses on basic NLP preprocessing: HTML→text, regex extraction, tokenization, normalization, stopwords, corpus statistics, POS tagging, stemming vs lemmatization, and NER.

Dataset (Webpage Used)

URL Used: https://www.fccollege.edu.pk/

This page is my university, where I did my bachelors it contains information about departments, academics, and university related different sections, so it is fine for my token statistics and pattern extraction tasks

P1T1: HTML to Plain Text Utility

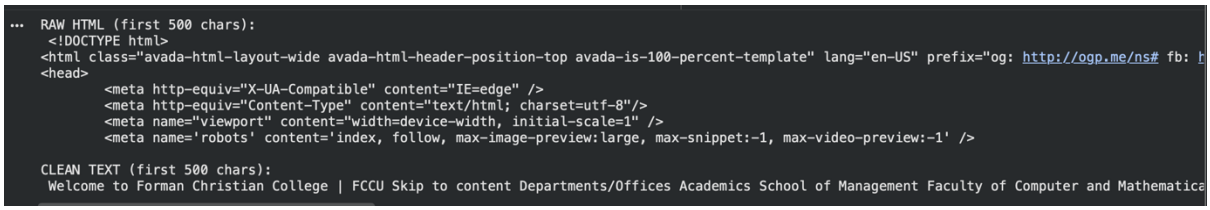
Goal

In this task, I created a function fetch_clean_text(url) to convert a webpage from HTML format into simple readable text. First, I downloaded the webpage

using urllib.request with a User-Agent header. Then, I used BeautifulSoup to remove HTML tags and extract only the visible text. I also removed extra spaces to make the text clean.

Output

- RAW HTML (first 500 chars):
- CLEAN TEXT (first 500 chars)



P1T2: Regex Extraction + Tokenization

Goal

The goal of this task was to extract email addresses and phone numbers from the cleaned text using regular expressions (regex) before tokenization, because tokenization can break email patterns. After extracting contact information, the text was tokenized into sentences and words.

Output

- **Unique emails found:** []
- **Unique phone numbers found:** ['92 42) 99231581']
- **Total sentences:** 19
- **Total tokens:** 1051

```
Email regex: [a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}
Phone regex: (?:\+?\d{1,3}[\s-]?)?(?:\(?\d{2,4}\)?[\s-]?)?\d{3,4}[\s-]?\d{3,4}

Unique emails: []
Unique phone numbers: ['92 42) 99231581']

Total sentences: 19
Total tokens: 1051
```

Explanation

Regex helps find patterns like emails/phones directly from raw text. After that, sentence tokenization splits text into sentences, and word tokenization splits it into tokens.

P1T3: Text Normalization

Goal

The goal of this task was to make the text clean, consistent, and easier to analyze. I converted all words to lowercase so that the same words are not treated differently due to capital letters. I removed extra spaces and line breaks to clean the text. I also replaced digits with the special token <NUM> so that different numbers are treated in the same way and do not increase the vocabulary size. Punctuation was handled carefully, keeping it only where necessary while focusing mainly on word tokens for analysis.

Output / Results

- **Before normalization (first 300 chars)**
- **After normalization (first 300 chars)**

```
BEFORE NORMALIZATION (first 300 chars):

Welcome to Forman Christian College | FCCU Skip to content Departments/Offices Academics School of
Management Faculty of Computer and Mathematical Sciences Computer Science Mathematics Statistics
Faculty of Education Education Health and Physical Education Faculty of Humanities English Mass
Communic

----------------------------------------------------------------------

AFTER NORMALIZATION (first 300 chars):

welcome to forman christian college | fccu skip to content departments/offices academics school of
management faculty of computer and mathematical sciences computer science mathematics statistics
faculty of education education health and physical edu
```

P1T4: Stopwords + Corpus Statistics

Goal

The goal of this assignment is to analyze the most frequent tokens in a text before and after removing stopwords, and to compare the results by reporting total token count, vocabulary size, and Type–Token Ratio (TTR) to observe changes in lexical diversity.

Results

Top-20 BEFORE stopwords (sample):
> (51), < (47), NUM (47), of (29), education (23), and (20), faculty (19), …

Top-20 AFTER stopwords (sample):
NUM (47), education (23), faculty (19), student (18), center (17), office (15), …

Statistics Summary

Metric	Before	After
Total Tokens	1156	774
Vocabulary Size	355	299
Type–Token Ratio (TTR)	0.3071	0.3863

```
P1T4 — STOPWORDS REMOVAL & CORPUS STATISTICS
================================================

Top-20 BEFORE Stopword Removal:

>                51
<                47
NUM              47
of               29
education        23
and              20
faculty          19
student          18
.                18
to               17
center           17
,                16
office           15
the              15
financial        14
fccu             12
university       11
forman           10
sciences         10
academic         10

------------------------------------------------

Top-20 AFTER Stopword Removal:

NUM              47
education        23
faculty          19
student          18
center           17
office           15
financial        14
fccu             12
university       11
forman           10
sciences         10
academic         10
aid              10
rector           9
students         9
services         8
college          7
health           7
alumni           7
programs         7

------------------------------------------------
```

```
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CORPUS STATISTICS:

Metric                          Before          After
------------------------------------------------------------
Total Tokens                    1156            774
Vocabulary Size                 355             299
Type-Token Ratio (TTR)          0.3071          0.3863

============================================================
```

Explanation

Stopwords (like *the, is, and, of*) are common but usually not meaningful for topic/keyword analysis. After removing them, vocabulary becomes more content-focused and TTR increases because the remaining words are more informative.

P1T5: POS Tagging & POS Distribution

Goal

The goal of this task is to perform Part-of-Speech (POS) tagging and identify and display the most frequent POS tags in the text, and extract example nouns and verbs to analyze the grammatical distribution of words.

Output / Results

Top 10 POS tags (by frequency):

- NN: 334
- JJ: 137
- NNS: 123
- VBP: 51
- NNP: 47
- VBG: 24
- VBD: 19
- VBZ: 10
- RB: 9
- VBN: 6

Example nouns (15):
['forman', 'college', 'fccu', 'skip', 'content', 'academics', 'school', 'management', 'faculty', 'computer', 'sciences', 'computer', 'science', 'mathematics', 'statistics']

Example verbs (15):
['urdu', 'pharmacy', 'faculty', 'fpsc', 'counseling', 'mercy', 'writing', 'student', 'fc', 'giving', 'alumni', 'chartered', 'dating', 'offers', 'university']

Explanation

POS tagging labels each word with its grammatical role (noun, verb, adjective, etc.). It helps in tasks like information extraction, parsing, and identifying important word types.

P1T6: Lemmatization vs Stemming

Goal

The goal of this task is to compare the effects of the Porter stemmer, Lancaster stemmer, and WordNet lemmatizer on a sample of 125 tokens to observe differences between stemming and lemmatization techniques

Output

Example rows from the produced table:

Token	Porter Stem	Lancaster Stem	Lemma
welcome	welcom	welcom	welcome
academics	academ	academ	academic
management	manag	man	management

```
================================================================================
P1T6 – STEMMING vs LEMMATIZATION COMPARISON
================================================================================
Token                Porter Stem          Lancaster Stem       Lemma
--------------------------------------------------------------------------------
welcome              welcom               welcom               welcome
forman               forman               form                 forman
christian            christian            christian            christian
college              colleg               colleg               college
fccu                 fccu                 fccu                 fccu
skip                 skip                 skip                 skip
content              content              cont                 content
academics            academ               academ               academic
school               school               school               school
management           manag                man                  management
faculty              faculti              facul                faculty
computer             comput               comput               computer
mathematical         mathemat             mathem               mathematical
sciences             scienc               sci                  science
computer             comput               comput               computer
science              scienc               sci                  science
mathematics          mathemat             mathem               mathematics
statistics           statist              stat                 statistic
faculty              faculti              facul                faculty
education            educ                 educ                 education
education            educ                 educ                 education
health               health               heal                 health
physical             physic               phys                 physical
education            educ                 educ                 education
faculty              faculti              facul                faculty
humanities           human                hum                  humanity
english              english              engl                 english
mass                 mass                 mass                 mass
communication        commun               commun               communication
philosophy           philosophi           philosoph            philosophy
religious            religi               religy               religious
studies              studi                study                study
urdu                 urdu                 urdu                 urdu
faculty              faculti              facul                faculty
natural              natur                nat                  natural
sciences             scienc               sci                  science
chemistry            chemistri            chem                 chemistry
physics              physic               phys                 physic
environmental        environment          environ              environmental
sciences             scienc               sci                  science
pharmacy             pharmaci             pharm                pharmacy
kauser               kauser               kaus                 kauser
abdulla              abdulla              abdull               abdulla
malik                malik                malik                malik
school               school               school               school
life                 life                 lif                  life
sciences             scienc               sci                  science
faculty              faculti              facul                faculty
```

Explanation

- Stemming cuts words roughly; it is fast but sometimes incorrect.
- Lemmatization returns dictionary form; it is more correct but needs linguistic rules.
- Lancaster is more aggressive than Porter, so it can shorten words too much.

P1T9: Named Entity Recognition (NER)

Goal

The goal of this task is to apply Named Entity Recognition to identify and extract named entities from the text under the categories of PERSON, ORGANIZATION, and GPE/LOCATION.

Output

- PERSON entities
- ORGANIZATION
- GPE/LOCATION

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P1T9 – NAMED ENTITY RECOGNITION (NER)
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PERSON Entities (Total: 0):
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--------------------------------------------------------------------------------

ORGANIZATION Entities (Total: 1):
--------------------------------------------------------------------------------
NUM

--------------------------------------------------------------------------------

GPE / LOCATION Entities (Total: 0):
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================================================================================
```

PART 2: Language Models and Smoothing

This part required training 4 models (unigram/bigram with smoothing) and evaluating perplexity on in-domain and out-of-domain corpora.

Dataset

- **Training:** Brown Corpus (NLTK)

- **Out-of-domain test:** Reuters Corpus (NLTK)

Preprocessing

- Added sentence markers: <s> at start and </s> at end (required).
- Lowercased tokens.
- Replaced rare words with <UNK> (threshold = 1) to reduce zero probability issues.

Training Summary

- Training sentences: **45,872**
- In-domain test sentences: **11,468**
- Out-of-domain test sentences: **54,716**
- Total training tokens: **1,020,759**
- Vocabulary size: **24,757**

Models Implemented

1. Unigram (unsmoothed)
2. Unigram (Laplace add-one smoothing)
3. Bigram (unsmoothed)
4. Bigram (Linear Interpolation smoothing)

"Unique / Good thing" I did here

- I computed perplexity in **log-space** (using log probabilities

Sentence Generation Files

Generated 20 sentences each and saved:

- unigram_output.txt
- smoothunigram_output.txt
- bigram_output.txt
- smoothbigramLI_output.txt

Perplexity

Perplexity was computed for both test sets.

Model	In-domain PP	Out-domain PP
Unigram (unsmoothed)	725.90	945.00
Unigram (Laplace)	728.50	943.83

Bigram (unsmoothed)	inf	inf
Bigram (Linear Interp)	275.44	666.99

```
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PART 2 — DATA SUMMARY
============================================================
Training sentences: 45872
In-domain test sentences: 11468
Out-of-domain test sentences: 54716

Example training sentence:
['<s>', 'he', 'let', 'her', 'tell', 'him', 'all', 'about', 'the', 'church', '.', '</s>']

============================================================
PART 2 — TRAINING STATISTICS (after <UNK>)
============================================================
Total training tokens: 1020759
Vocabulary size: 24757

Top 10 most common words:
the             55897
,               46676
<s>             45872
</s>            45872
.               39479
of              29092
and             23140
to              20991
<UNK>           20399
a               18659

============================================================
PART 2 — MODELS TRAINED
============================================================
1) Unigram (unsmoothed)
2) Unigram (Laplace)
3) Bigram (unsmoothed)
4) Bigram (Linear Interp)

Saved 20 generated sentences to:
 - unigram_output.txt
 - smoothunigram_output.txt
 - bigram_output.txt
 - smoothbigramLI_output.txt

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PART 2 — PERPLEXITY RESULTS
============================================================
Unigram (unsmoothed)      | In-domain PP: 725.90 | Out-domain PP: 945.00
Unigram (Laplace)         | In-domain PP: 728.50 | Out-domain PP: 943.83
Bigram (unsmoothed)       | In-domain PP: inf | Out-domain PP: inf
Bigram (Linear Interp)    | In-domain PP: 275.44 | Out-domain PP: 666.99
```

Answers to Required Questions (Part 2)

Q1) In unigram, what controls sentence length? How is it different from bigram?

- In unigram generation, each word is chosen independently. Sentence ends when </s> is sampled, so the probability of **</s>** controls average sentence length.
- In bigram generation, the next word depends on the previous word, so endings happen in a more context-based way and produce more structured sentences.

Q2) Do models assign drastically different probabilities? Why?

- Yes. Unsmoothed bigram often gives zero probability for unseen bigrams, making full sentence probability become zero (and perplexity becomes infinity).
- Smoothed models avoid this by giving small probabilities to unseen events.

Q3) Which model produces better / realistic sentences?

- The Smoothed Bigram (Linear Interpolation) produces more sentences than unigram because it uses context which is previous word and smoothing prevents dead ends.

Q4) Which test corpus has higher perplexity and why? (include values)

- Out-of-domain test has higher perplexity in working models because language and vocabulary style differs from training domain.
  Example:
- Unigram unsmoothed: **945.00 (out) > 725.90 (in)**
- Bigram LI: **666.99 (out) > 275.44 (in)**

PART 3: Byte Pair Encoding for Urdu

This part does the BPE for Urdu segmentation and calculates the average characters per token for a self-collected Urdu corpus.

P3T1: BPE for Urdu Word Segmentation

Goal

Segment 50 Urdu sentences without spaces into meaningful subwords/words using BPE and a dictionary, and save the output.

Results

- Dictionary words: **5691**
- Sentences: **50**
- BPE merges learned: **500**
- Output saved: **50_segmented.txt**

```
Dictionary words: 5691
Sentences: 50
BPE merges learned: 500
✅ Saved output: 50_segmented.txt
```

P3T2: Average Characters per Word in Urdu Text

Goal: The goal of this task is to collect a large Urdu corpus at least 10,000 tokens, but I collected 12k+ tokens and then tokenized the text, calculated the total number of tokens and the average number of Unicode characters per token.

Urdu Corpus Sources

1. BBC Urdu live page: https://www.bbc.com/urdu/live/ckg16vlk67nt?page=3
2. UrduPoint kids story: https://www.urdupoint.com/kids/detail/moral-stories/ajnabi-sardar-2432.html

Results

- **Total tokens:** 12,654
- **Average Unicode characters per token:** 3.637

```
Total tokens: 12654
Average Unicode characters per token: 3.637
```

PART 4: Embedding Models

This part covers embedding models.

Datasets Used

- **Primary (unlabeled):** Reuters news-style dataset (4,000 docs used in notebook output)
- **Secondary (labeled):** AG News (train 120,000 / test 7,600)

P4T1: TF-IDF

Goal

The goal is to implement TF-IDF from scratch and report vocabulary size, sparsity ratio, OOV rate, and cosine similarity, while comparing results for word-level tokens vs BPE tokens.

Results

Word-level TF-IDF

- Vocab: **15000**
- Sparsity: **0.9817**
- OOV: **0.0343**
- Avg cosine similarity: **0.1651**

BPE-based TF-IDF

- Vocab: **186**
- Sparsity: **0.1067**
- OOV: **0.0**
- Avg cosine similarity: **0.8419**

```
----- P4T1 TF-IDF (FAST) -----
Vocab word: 15000 Sparsity: 0.9817 OOV: 0.0343
Vocab bpe : 186 Sparsity: 0.1067 OOV: 0.0
Avg cosine word: 0.1651
Avg cosine bpe : 0.8419
```

Analysis

BPE reduces vocabulary and almost removes OOV, so document vectors become less sparse.

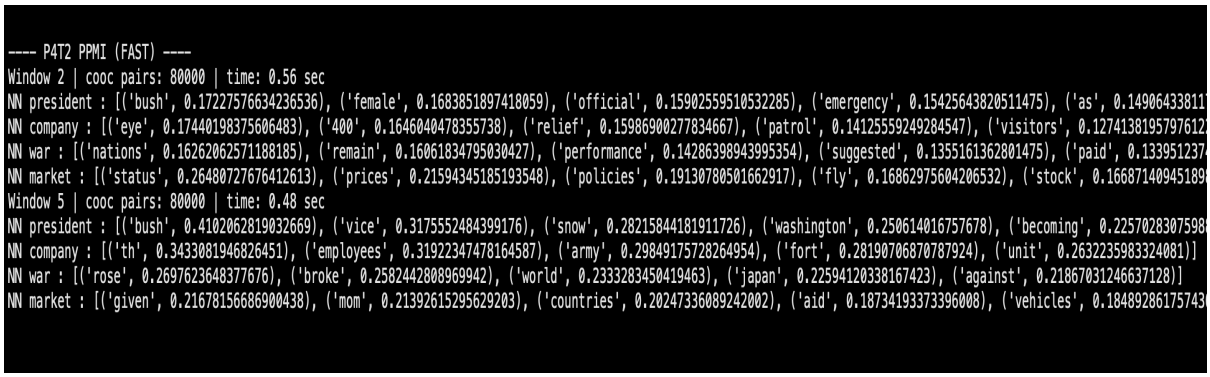
P4T2: Count-Based Word Embeddings

Goal

Build a word-context co-occurrence matrix, apply PPMI, test with different windows, show nearest neighbors.

Results

- Window 2 co-occ pairs: **80000** (fast run)
- Example nearest neighbors shown for a word like **“president”** (neighbors printed in notebook)



Explanation

PPMI highlights word-context relationships by down-weighting very common contexts and emphasizing informative co-occurrences.

P4T3: Neural Word Embeddings

Goal

The goal of this task is to implement SGNS (Word2Vec-style) from scratch without using embedding libraries, and report the training loss**,** training time**,** nearest neighbors**,** and a brief coverage/OOV discussion**.**

Results

- Device: CPU
- Vocab: **6000**
- Pairs: **40000**
- Epoch 1 loss: **4.1631**
- Train time: **0.61 sec**
- Example neighbors printed for “president”

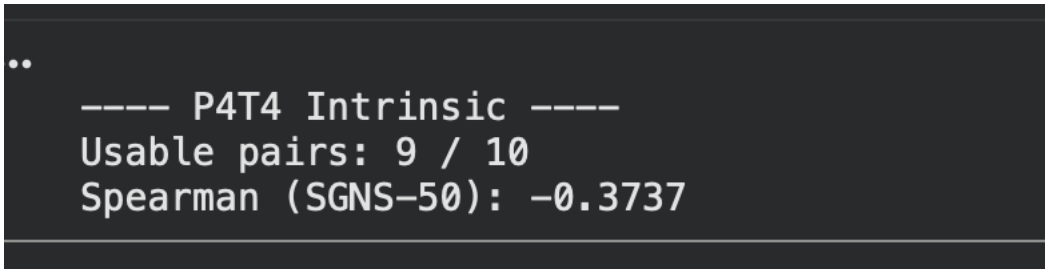
P4T4: Intrinsic Evaluation

Goal

The goal is to evaluate intrinsic embedding quality using custom word similarity pairs.

Results

- Usable pairs: **9 / 10**
- Spearman (SGNS-50): **-0.3737**



P4T5: Sentence Embeddings

Goal

The goal is to create sentence vectors using mean pooling and a TF-IDF weighted average**,** then evaluate their performance

Results

- Mean pooling accuracy: **0.51**
- TF-IDF weighted accuracy: **0.5183**

```
---- P4T5 Sentence similarity ----
Mean pooling accuracy: 0.51
TFIDF-weighted accuracy: 0.5183
```

P4T6: Extrinsic Evaluation

Goal

The goal is to use the learned embeddings or feature representations for a downstream task and evaluate accuracy and F1 score.

Results (AG News Classification)

- **TF-IDF Accuracy:** 0.8307
- **TF-IDF Macro-F1:** 0.8282
- **Sentence(mean) Accuracy:** 0.4473
- **Sentence(mean) Macro-F1:** 0.4407

```
---- P4T6 Extrinsic (FAST) ----
TFIDF Accuracy: 0.8307 Macro-F1: 0.8282
Sentence(mean) Accuracy: 0.4473 Macro-F1: 0.4407
```